

Introduction to the Internet of Things, Reference Architecture and Overview of Communication Technologies

Communication Technologies for the Internet of Things:
Lesson 01 Course Notes

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Abstract

This document gathers, in continuous prose, the content of Lesson 01 (theoretical) of the course Communication Technologies for the Internet of Things. It is organised in three parts. The first presents the course, the IoT reference architecture, the machine to machine communication paradigm and signal conversion (chapters 1 to 4). The second covers the fundamentals of communication networks: Internet architecture, packet switching and performance, the layered communication model, access to shared transmission media and cellular mobile networks, ending with the bridge to IoT technologies (chapters 5 to 9). The third goes through the overview of communication technologies available for IoT solutions, with particular emphasis on short

range wireless solutions (chapters 10 to 13). It is intended for independent study and complements the slides used in class.

1 Course Presentation

This chapter frames the course Communication Technologies for the Internet of Things: its objectives, the syllabus, the assessment regime and the reference bibliography.

1.1 Objectives and Syllabus

By the end of the course, students should be able to describe and apply the concepts required to define the architecture and design Internet of Things (IoT) applications, select and apply the standards and communication technologies best suited to the requirements and constraints of a specific solution, select and apply the standards and data distribution technologies best suited to the scenario at hand, and design and implement IoT applications using communication solutions and message oriented middleware (MoM).

The syllabus is organised in six blocks. The first covers the IoT reference architecture. The second addresses the machine to machine communication paradigm. The third covers wide area network (WAN) access technologies, with emphasis on low power networks, known as LPWAN. The fourth block studies network technologies specific to IoT (IPv4, IPv6, 6LoWPAN, 6TiSCH, Neighbor Discovery, RPL, CoRPL and CARP) as well as next generation IP protocols. The fifth block presents the session layer protocols used in IoT: HTTP, CoAP, XMPP, AMQP and MQTT. The sixth and final block focuses on Message-Oriented Middlewares, the mechanism that allows the data collected by devices to be distributed and processed.

1.2 Assessment and Workload

Assessment follows a discrete model, split across two moments of equal weight. The theoretical component is a written exam taken in person, worth 50% of the final grade. The practical component is an integrative mini-project, also worth 50%, delivered in the last lesson of the semester. Students who do not pass by this route, or who wish to improve their grade, are assessed in the Resit or Special period through a single exam covering both the theoretical and the practical content of the course.

The workload of the course is measured in ECTS credits, European Credit Transfer and Accumulation System, a reference unit valid across the European space, defined under the Bologna Process. Each credit corresponds to about 27 hours of student work, between contact hours and independent study. A typical semester amounts to 30 ECTS; this course has 6 ECTS, which corresponds to approximately 144 hours of work over the semester. The 3 weekly contact hours correspond to about 5 hours of independent work, which is why the content and the practical guides are prepared with that work outside the classroom in mind.

1.3 Integrative Mini-Project

The mini-project consists of developing a complete IoT application, from the sensor to the message broker. It must integrate at least one of the short range communication technologies studied in the course (NFC, BLE or Zigbee) together with a data distribution mechanism based on a Message-Oriented Middleware. Development runs throughout the semester, with direct support in the practical lessons, and the demonstration and delivery take place in the last lesson. The detailed specification, with the concrete requirements of each topic and the physical quantities to monitor, is provided in a separate document.

1.4 Bibliography

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2 Fundamentals of the Internet of Things

This chapter introduces the fundamental concepts of the Internet of Things: its definition, the origin of the term, the hardware platforms most used for prototyping, and the reference model that places each technology within a wider architecture.

2.1 Definition and Origin of the Concept

The Internet of Things designates the set of physical objects equipped with sensors, processing capability and connectivity, able to exchange data with each other and with systems on the network without the need for constant human intervention. The expression *Internet of Things* is generally attributed to Kevin Ashton, who used it in 1999 in the context of the Auto-ID Center at the Massachusetts Institute of Technology, associating it initially with radio frequency identification (RFID) as the enabling technology. Cisco places the effective start of IoT (understood as the moment when the number of devices connected to the Internet exceeded the number of people on the planet) between 2008 and 2009.

The continuous growth in the number of connected devices brings with it a problem of scale: the collection, aggregation and analysis of data volumes that grow far faster than the human capacity to interpret them manually. This phenomenon is directly associated with the concept of Big Data, and accounts for much of the architecture and protocols studied throughout this course.

2.2 Prototyping Platforms

The development of IoT solutions almost always begins with a hardware prototype. Throughout the practical lessons, the reference platform is Arduino, an open source hardware and software ecosystem made up of single board microcontrollers and the corresponding development environment. Arduino allows sensors to be read and actuators to be controlled without requiring digital electronics built from scratch; the hardware is distributed under a Creative Commons licence and the software under GNU LGPL or GPL licences, which allows any manufacturer to produce boards compatible with the original ecosystem.

Other families of microcontrollers exist, with distinct characteristics. Teensy offers processors more powerful than the classic Arduino on small boards. The MSP430, from Texas Instruments, is aimed at applications with very low energy consumption. NodeMCU integrates the ESP8266 Wi-Fi module directly into the microcontroller, which simplifies direct connection to the local network. The STM32, from STMicroelectronics, is a family of 32 bit microcontrollers used when more performance and more peripherals are needed. PocketBeagle, from the BeagleBoard family, occupies an intermediate position between a microcontroller and a single board computer.

When the solution requires a complete operating system, for instance to act as a gateway between short range devices and the network, the reference platform is the Raspberry Pi, a family of single board computers developed in the United Kingdom by the Raspberry Pi Foundation, with the original purpose

of promoting the teaching of computing in schools. Unlike a microcontroller, the Raspberry Pi runs a complete operating system, which makes it easier to develop more complex applications and to integrate directly with IP networks. In this course, the Raspberry Pi will take the role of gateway in several practical lessons. Alternatives exist on the market, such as the ASUS Tinker Board, the NVIDIA Jetson Nano, the Onion Omega2+, the BeagleBone Black and the Orange Pi, each with a different balance between price, performance and energy consumption.

2.3 The Internet of Things Reference Model

To place each technology studied within a coherent architecture, this course uses as reference the IoT World Forum Reference Model, proposed by Cisco in 2014. The model organises an IoT solution into seven layers, from the physical capture of data to its use by business processes.

The first layer corresponds to the physical devices and controllers, the sensors, the actuators and the other objects that are the origin and the final destination of the information. The second layer, connectivity, comprises the networks and communication protocols that carry data between devices, gateways and the wider network; it is to this layer, and to the previous one, that most of the syllabus of this course is devoted. The third layer corresponds to edge computing, or edge/fog computing: processing carried out near the origin of the data, which filters, converts or reacts to events before sending them to the network. The fourth layer, data accumulation, deals with the storage of the collected data, turning event streams into persistent data sets.

The fifth layer, data abstraction, aggregates and normalises data coming from multiple sources, making it accessible to applications in a consistent way. The sixth layer, application, interprets that data for concrete purposes, such as reporting, analysis or automatic process control. The seventh and final layer, collaboration and processes, integrates the information into business

processes and into decision support for people and organisations. The upper layers of the model relate directly to the Message-Oriented Middlewares and the data distribution protocols studied in the final part of the syllabus.

3 The Machine to Machine Paradigm

This chapter presents the machine to machine communication paradigm, historically prior to the Internet of Things, and discusses the similarities and differences between the two paradigms.

3.1 Definition and M2M Area Networks

Machine-to-Machine, or M2M, designates the interconnection of machines or devices for the purpose of remote monitoring, control and data exchange, without direct human intervention. An M2M area network is made up of nodes (the machines themselves) equipped with embedded sensing, actuation and communication modules. Protocols used in these local networks include ZigBee, Bluetooth, Modbus, M-Bus, Wireless M-Bus, Power Line Communication (PLC), 6LoWPAN and IEEE 802.15.4, protocols that will be revisited in detail in the lessons dedicated to network technologies.

The network that interconnects different M2M area networks with each other is typically IP based, whether over wired or wireless media. This distinction matters: inside the M2M area network, proprietary or non IP protocols predominate; outside it, on the wider communication network, IP predominates.

3.2 M2M Gateways

Since M2M area networks often use proprietary or non IP protocols, the nodes of one network cannot, by themselves, communicate directly with nodes of another M2M network. The M2M gateway solves this limitation: it translates between the local protocol of the M2M area network and the IP protocol used

in the communication network, allowing data originating in a sensor network to reach remote systems. In this course, that role will be played, in different practical lessons, by a Raspberry Pi configured as a gateway.

3.3 IoT and M2M: Convergences and Differences

The two paradigms share similar goals, monitoring, controlling and exchanging data between devices, but they differ in several respects.

As to communication protocols, M2M relies mostly on proprietary or non IP protocols inside the local area network. IoT may also use non IP protocols in sensor networks, but tends to expose the data through IP protocols as soon as it leaves the sensor network.

As to homogeneity, M2M networks typically have homogeneous machine types within the same area network. In IoT, the term “thing” refers to any physical object with a unique identifier, able to sense and communicate with the environment and with other applications, which results, in practice, in greater heterogeneity of devices within the same solution.

As to technological emphasis, M2M rests mostly on embedded hardware modules, while IoT gives more weight to software, to the integration of data from multiple sources and to the analysis of that data.

As to data collection and analysis, in M2M the data is typically collected and stored in local infrastructure, tied to point solutions and to a single supplier. In IoT, the data is collected and processed mostly in the cloud, whether public, private or hybrid, which facilitates scale and integration between distinct solutions.

As to applications, M2M applications are normally local: diagnostics, service management, business applications installed on the premises themselves. IoT applications reside in the cloud and include advanced analytics, remote diagnostics and integration with wider enterprise systems.

4 Sensors and Signal Conversion

This chapter deals with the origin of the data in an IoT solution: the sensors, and the process that converts their readings, essentially analogue, into a format that digital systems can process.

4.1 Sensors as a Source of Data

The term “thing”, as used in IoT, represents in abstract terms any physical element able to produce or consume information in a solution, typically a sensor or an actuator. A sensor may be simple, dedicated to monitoring a single physical quantity, such as temperature or humidity. It may also be complex, combining several quantities to infer behaviour, for instance the health state of a person or an animal, or patterns of social and financial activity. The choice between a simple and a complex sensor directly constrains the communication and processing requirements of the whole solution: more quantities monitored means, in general, more data to transmit and more processing capability required, whether on the device itself or on a nearby gateway.

4.2 Analogue to Digital and Digital to Analogue Conversion

Microprocessors are digital machines: they operate on discrete values, represented internally in binary. The physical world observed by sensors is, by contrast, essentially analogue, with quantities such as temperature, pressure or sound varying continuously in time. An analogue to digital converter (*Analog-to-Digital Converter*, or ADC) is therefore needed to transform the continuous signal coming from the sensor into a numeric sequence that the processor can handle. When, conversely, the digital system needs to act on the physical world, for instance to control the speed of a motor, the reverse process is required, carried out by a digital to analogue converter (*Digital-to-Analog Converter*, or DAC).

4.3 Sampling and the Nyquist Theorem

Sampling consists of measuring the input signal periodically, producing a sequence of values that represents it over time. The central question of sampling is how often to measure: if the sampling frequency is too low, the resulting sequence of samples does not allow the original signal to be correctly reconstructed, and information is lost irrecoverably.

The Nyquist Theorem, formulated by Harry Nyquist, answers this question: the sampling frequency must be at least twice the highest spectral component present in the input signal. Formally, $f_s = 2B$, where B represents the highest spectral component of the signal, that is, the fastest variation present in the signal to be captured. In practice, this minimum condition is rarely sufficient: a low pass filter is used to bound B to a known value, and a sampling frequency five to ten times higher than that maximum component is adopted, so as to provide a safety margin against filter imperfections and noise.

4.4 Quantisation Error and Resolution

Besides the sampling frequency, analogue to digital conversion involves a second limitation: the amplitude of the signal must also be represented by a finite number of discrete values. This limitation introduces what is called quantisation error, whose magnitude depends directly on the number of bits used in the conversion. With 3 bits of resolution only 8 distinct levels are obtained (2^3), which produces a coarse representation of the original signal. With 4 bits, 16 levels are obtained (2^4), with a noticeably better approximation. The larger the number of bits, the smaller the quantisation error, but also the larger the volume of data produced, with direct implications for the bandwidth needed to transmit it.

Some practical examples illustrate this trade-off. In classical telephony, the signal is sampled at 8 kHz, with 8 bits per sample, which gives a transmission

rate of 64 kbit/s; one minute of conversation takes about 469 KB. In CD audio, sampling is done at 44.1 kHz, with 16 bits per sample and two channels, which gives a rate of about 1411 kbit/s; one minute of music takes about 10.5 MB, and a CD with 74 minutes holds about 740 MB. Other common sampling rates include the 48 kHz used in DAT, DV and DVD-Video, and the 96 kHz used in DVD-Audio; in terms of resolution, low quality applications use 8 bits, CD and DV use 16 bits, DVD uses 24 bits, and the internal representation in software often uses 32 bit floating point.

These parameters (sampling frequency and number of resolution bits) will come up again whenever the design of a data acquisition system in IoT is discussed, because together they determine the quality of the reconstructed signal and the volume of data that the communication network has to carry.

5 Architecture of Communication Networks

The previous chapters dealt with the origin of the data and with the model that organises an IoT solution into layers. This chapter, and the three that follow, deal with what happens between the sensor and the application: the communication network itself. The order is not arbitrary. Practically every technology studied in the rest of the semester, from Bluetooth Low Energy to NB-IoT, is a different answer to the same set of fundamental problems, and it is more economical to understand those problems once, in the abstract, than to rediscover them technology by technology.

5.1 Components and Structure of the Internet

A communication network is made up of three kinds of element. **Hosts** are the end devices that run applications, typically organised into clients and servers. **Intermediate equipment**, routers and switches, relays data already divided into smaller units. **Links**, in optical fibre, copper, radio or satellite, join them to each other, each with an available bandwidth and with characteristics of its

own. A set of these elements managed by a single administrative entity constitutes a **network**; the Internet is the result of interconnecting thousands of independent networks, belonging to operators, homes, companies and educational institutions.

This description applies in full to an IoT solution. Sensors and actuators are hosts, the gateway is the intermediate equipment, and the local sensor network is just one more network to interconnect. The difference lies in the constraints, not in the nature of the problem.

Architecturally, the Internet divides into three zones. At the **edge** are the hosts, with servers typically concentrated in data centres operated by cloud service providers. The **access networks** are the links, wired and wireless, that join end devices to the first routers of the network. The **core** is the mesh of interconnections between networks, the network of networks itself. The division is useful because each zone solves a different problem and therefore admits different technologies. In an IoT solution, the sensors occupy the edge, the short range wireless technology is the access network, and the gateway is the point of passage to the core. It is worth noting that the vast majority of technological innovation in IoT happens in the access network: that is where range, consumption and cost per device are decided.

At the top of this structure, Internet service providers interconnect with each other in a hierarchy. At the centre are tier 1 operators, with global presence, alongside large content providers with networks of their own. Internet Exchange Points allow distinct networks to exchange traffic directly, without crossing a third operator. The resulting network is extremely complex, and the path between two points is rarely the geographically shortest one.

5.2 Protocols and Standardisation

A **protocol** is a set of rules defining the order and format of the messages exchanged, as well as the actions to take upon receiving each message.

Examples are the messages and actions needed for a device to associate with a network, or to fetch a page from a Web server. The distinction from the physical medium is essential: without a common protocol, two pieces of equipment connected to the same medium remain unable to communicate. The physical link is a necessary condition, but not a sufficient one.

Standards ensure that implementations from different manufacturers interoperate, which is particularly decisive in a market as fragmented as that of IoT. The IETF (*Internet Engineering Task Force*) publishes specifications of protocols, mechanisms and good practice in the form of RFCs. Other bodies relevant to this course are the IEEE, responsible for the standards 802.3 (Ethernet), 802.11 (Wi-Fi) and 802.15.4 (low rate personal area networks), the 3GPP, responsible for the standardisation of cellular networks, and industry alliances such as the Bluetooth SIG, the Connectivity Standards Alliance and the LoRa Alliance.

5.3 Types of Access Network

Residential access today relies on two main families. **Hybrid fibre and coaxial cable networks**, standardised in EuroDOCSIS, and **fibre networks** of the GPON type (*Gigabit Passive Optical Network*). Both share two characteristics: they are asymmetric, offering higher throughput downstream than upstream, and they are shared, in the sense that several homes compete for the same capacity up to the operator aggregation point. **Institutional access** combines wired and wireless links, interconnecting switches and routers of companies or educational institutions, typically with a high speed link to the operator.

As to wireless access, two scales should be distinguished. **Wireless local area networks** (WLAN) operate inside or near buildings, with ranges around 50 metres and throughput that today exceeds 1 Gbit/s. **Cellular or mobile networks** are provided by operators, with ranges of the order of kilometres and throughput that can likewise reach 1 Gbit/s in the most recent

generations. Finally, **data centres** use very high capacity links, of 100 Gbit/s or more, between large numbers of servers.

The choice of access network is the most structural architectural decision in an IoT solution, because it simultaneously constrains range, consumption, cost per device and the operating model.

5.4 Transmission Media

Transmitting data between two points requires, first, converting that data into physical signals the medium can carry. In **guided media**, signals propagate along a fixed physical support. Twisted pair cable (UTP) carries electrical signals and is classified into categories (Cat 5e, Cat 6, Cat 6a). Optical fibre carries optical signals and is classified by mode of operation, single mode (OS1, OS2) or multimode (OM3, OM4). Coaxial cable, with two concentric conductors, carries electrical signals and is classified by diameter and impedance (RG-6, RG-58). Guided media offer predictability and immunity to interference, but impose fixed infrastructure, incompatible with mobile devices or with dispersed installations.

In **unguided media**, signals propagate freely in space, in the form of radio or optical waves. The decisive consequence is that the medium becomes shared: everything transmitted on a given frequency in a given area is received by all devices within range. From this follows **half-duplex** operation, that is, a device does not transmit and receive at the same time on the same frequency, because its own signal dominates the receiver. The medium also introduces effects that guided media do not have: reflections off surfaces, obstruction by obstacles and noise from other sources. Unguided technologies include Wi-Fi, cellular networks, Bluetooth, ZigBee, Z-Wave, LoRa, microwave, satellite and optical laser links.

This distinction between guided and unguided media is the source of nearly all the differences in behaviour between the technologies studied in this course,

and chapter 7 revisits it in detail when dealing with the problem of accessing a shared medium.

6 Switching, Performance and the Layered Model

This chapter deals with two topics that at first sight have little in common: how information crosses the network and what delays it suffers along the way, and how protocols are organised among themselves. Both are nonetheless indispensable for understanding the design choices of IoT technologies, which are to a large extent attempts to work around the costs identified here.

6.1 Packet Switching

Messages are not transmitted whole. They are divided into **packets**, each with a maximum size, and each packet carries the information needed to determine its destination. Routing protocols build, in each router, tables indicating the best outgoing link for each destination, and the network forwards each packet individually, from source to destination.

Two mechanisms constrain the performance of this process. The first is **transmission latency**: a packet of L bits on a link of R bit/s takes at least L/R seconds to be transmitted on each link. The second is the **store and forward** principle, under which the packet must arrive in full at the router before being retransmitted on the next link. The combination of the two makes the delay accumulate at every hop. In an IoT mesh network with several hops, such as a ZigBee network, this effect is directly responsible for the end to end delay observed, and explains why the number of hops is a design parameter and not an implementation detail.

When the traffic arriving at a piece of equipment exceeds the capacity of the outgoing link, packets wait in **queues**. Links with lower transfer rates therefore increase the latency observed, and not only the transmission time. If the queue fills up, arriving packets are simply discarded: this is the main source of packet loss in a network. In IoT solutions, the bottleneck is typically at the gateway, where the aggregate traffic of dozens of sensors has to fit into a single outgoing link.

6.2 Performance Metrics

The performance of a network is characterised by three metrics. **Latency** is the sum of all delays a packet suffers from source to destination. **Packet loss** is the fraction of packets discarded, typically through exhaustion of memory in the queues. **Throughput** is the effective rate at which bits reach the receiver. The three are not independent, and none of them is optimised without cost to the others.

Latency breaks down into four components, summed along each network segment. The **processing delay** is the time spent handling the packet in the router, including the forwarding decision and error checking, and is typically below a microsecond. The **queuing delay** is the time the packet waits before it can be transmitted; it depends on the level of congestion and is the most variable component of all. The **transmission delay**, given by L/R , is the time needed to place all the bits of the packet on the link. The **propagation delay**, given by d/s , is the time the signal takes to physically travel the link, where d is the link length and s the propagation speed in the medium.

The distinction between the last two components is essential and frequently confused: the transmission delay depends on the throughput of the technology, whereas the propagation delay depends only on distance and on the physics of the medium. Raising the throughput of a link reduces the first and does not affect the second.

As to throughput, a distinction is made between the instantaneous rate, which is the rate at which the receiver gets data at a given moment, and the average rate, which is the volume transferred divided by the total time. On a path with several links in series, throughput is limited by the slowest link, known as the **bottleneck**. In IoT, the bottleneck is almost always on the first link, the one joining the sensor to the gateway, and not in the network core. Two tools allow these quantities to be measured in practice: *ping*, which measures end to end latency and the loss rate, and *traceroute*, which measures latency hop by hop along the path.

6.3 Layered Organisation

Protocols are organised into **layers**, each solving a well delimited part of the overall communication problem. Each layer offers a service to the layer immediately above and uses the services of the layer immediately below, without knowing its internal workings. This independence allows **substitutability**: the implementation of one layer can be replaced without changing the others, provided the service interface is preserved. The principle is the same one that organises an air journey, the traditional postal service or admission to a hospital emergency department, that is, successive stages, each with its own responsibility.

It is thanks to this principle that the same IoT application runs over Wi-Fi, over Ethernet or over a cellular network, with no change to the code. It is also why it makes sense to study medium access in the abstract, as will be done in the following chapter: the conclusions apply to any technology occupying that layer.

The **TCP/IP model** organises communication into five layers. The **application** layer offers application oriented services, such as HTTP, SMTP, IMAP or FTP; in IoT, this is where MQTT, CoAP and AMQP live, studied in the final lessons of the course. The **transport** layer identifies the applications on each device and

handles segmentation, flow control and retransmission, with the TCP and UDP protocols. The **network** layer identifies devices and forwards packets across the Internet, with fragmentation where needed, through the IP protocol. The **link** layer encapsulates data into frames, detects errors and handles transmission between directly connected devices, and includes Ethernet, Wi-Fi, EuroDOCSIS and GPON. The **physical** layer converts bits into radio, electrical or optical signals, and defines the physical interface, with connectors and cables. The two lower layers are the central object of this course, and are where IoT technologies depart most from conventional solutions.

The **OSI model** is later than the TCP/IP model and breaks the application layer into three distinct layers, seven in total. The **presentation** layer specifies the interpretation of the data, including compression and encryption. The **session** layer, a session being a set of communication channels, allows sessions to be resumed after failure and handles authentication. The remaining layers correspond in practice to those of the TCP/IP model. The OSI model is widely used as reference vocabulary, even though the real Internet follows the TCP/IP architecture; the designation “session layer protocols”, which gives its name to Lesson 11 of this course, comes precisely from this model.

6.4 Encapsulation and its Cost in IoT

At each layer, the data received from above is wrapped in a header of its own before being handed to the layer below. The application message becomes a **segment** at the transport layer, a **datagram** at the network layer and a **frame** at the link layer. At the destination, the reverse process removes each header in turn, until the original message is delivered to the application. Intermediate equipment processes only up to the layer it needs: a switch decapsulates up to the link layer, a router up to the network layer.

Every header added represents, however, data that occupies space on the medium and energy to transmit, without carrying any useful application

information. In conventional networks, this cost is negligible compared with the volume of the data. In IoT, where the useful information may be a single temperature value of a few bytes, headers can exceed the content itself. It is this observation, and not an aesthetic preference, that motivates 6LoWPAN, studied in Lesson 09, whose essential function is to compress IPv6 headers down to sizes compatible with the frames of a sensor network.

7 The Link Layer and Medium Access

This chapter deals with the central problem of any wireless technology and therefore of practically every technology studied in this course: how several devices share a transmission medium without cancelling each other out. It is the longest chapter of this part, and not by accident: the differences between Bluetooth, ZigBee, Wi-Fi, LoRa and NB-IoT lie, to a large extent, in the answers each gives to this problem.

7.1 The Data Link Layer

The link layer establishes the communication channel between **adjacent nodes**, a node being any end device, router or switch. Its responsibility is to transfer frames, which contain the network layer datagrams, between directly connected nodes. It has a feature that sets it apart from all the others: it is the only layer whose protocol may change at every hop. A datagram can cross Wi-Fi, then Ethernet, then fibre, without ever being altered.

Links divide into two types. **Point to point links** are terminated directly between two devices, as in a modern switched Ethernet link; they operate in full-duplex and require no coordination. **Shared links with broadcast** connect multiple devices to the same medium, as in Wi-Fi, satellite, cable or GPON; they operate in half-duplex and necessarily require a coordination protocol, called *Medium Access Control* (MAC). Practically every wireless IoT

technology is a shared link, which means that medium access control is not, for them, an optional detail.

The link layer offers four services. **Framing** adds the link layer addresses and the fields needed for additional functionality. **Access to the transmission medium** defines the rules for when a frame may be sent, and may be absent on point to point links in full-duplex. **Reliable delivery between adjacent nodes** is used on links with many errors; in Wi-Fi, for instance, every frame is acknowledged by the receiver. **Error detection and correction** identifies frames corrupted by the medium, typically through a check code.

7.2 The Medium Access Problem

On a shared medium, if two devices transmit at the same time, the signals overlap and interfere with each other. The result is a **collision**: neither transmission is correctly received, and both are wasted. What makes it worse, on a radio medium, is that the collision happens at the **receiver**, and is therefore unknown to the transmitters at the moment it occurs. It is this asymmetry between where the decision is taken and where the problem shows up that makes medium access a genuinely difficult problem.

A medium access protocol is therefore a **distributed algorithm**: each device decides when to transmit using only local information. A good protocol should approach the full channel throughput when only one node transmits, share capacity fairly when several transmit, and work without central coordination. In IoT a further criterion applies, one that conventional networks typically ignore: every collision is energy spent with no result at all, and energy is the scarcest resource on the device.

Multiple access protocols fall into three families. In **channel partitioning**, the channel is divided into portions (of time, of frequency or of code) and each device uses its own exclusively. In **taking turns**, devices transmit in turn, coordinated by polling or by the circulation of a token. In **random access**,

each device transmits when it has data, accepting the risk of collision and resolving it when it occurs. None of these families is universally superior, and the choice depends on the traffic load, the number of nodes and the tolerance to delay.

7.3 Channel Partitioning

In **TDMA** (*Time Division Multiple Access*), time is divided into frames, and each frame into slots assigned to specific devices. Each device transmits only in its own slot, at the full speed of the channel, and stays silent during the others. The advantages are the complete absence of collisions, a deterministic maximum delay and, above all in IoT, the possibility of switching the radio off outside its own slot, with a direct energy gain. The disadvantages are the wasted slot when the device has no data to send, and the need for strict time synchronisation between all nodes.

In **FDMA** (*Frequency Division Multiple Access*), the available band is divided into sub-bands, and each device transmits continuously in its own. As in TDMA, there are no collisions, but each device is limited to a fraction of the total throughput, even when the others are idle. It also requires guard bands between sub-bands to avoid mutual interference, which consumes spectrum without carrying data. It was the technique of the first analogue cellular networks and remains present, combined with others, in practically every current system.

In **CDMA** (*Code Division Multiple Access*), all devices transmit in the same band at the same time, but each multiplies its data by a distinct code. If the codes are mutually orthogonal, the receiver can separate the transmission of interest, treating the others as noise. CDMA is robust to narrowband interference and does away with the strict time coordination of TDMA, but requires tight power control, otherwise the nearest transmitters drown out the more distant ones.

OFDMA (*Orthogonal Frequency Division Multiple Access*) divides the band into many narrow subcarriers, mutually orthogonal. Each device is assigned a set of subcarriers during a set of time intervals, combining the logic of FDMA and TDMA into a two dimensional grid. This flexibility allows the capacity assigned to each device to be adjusted with fine granularity, and is the technical basis of 4G, 5G and Wi-Fi 6.

7.4 Taking Turns

In **polling**, a central node asks each device in turn whether it has data to transmit, and grants it the floor if so. It removes collisions and allows the peripheral devices to be very simple, but introduces the delay of the polling itself and creates a single point of failure. This is exactly the model of classic Bluetooth, in which the master device polls the slaves of the piconet, as will be seen in Lesson 04.

In **token passing**, a special frame circulates among the devices, granting the right to transmit to whichever holds it. It requires no central node and guarantees fair access, but losing the token paralyses the network until a recovery mechanism regenerates it.

Both approaches are efficient under high load, when practically every node always has data to send, and inefficient under low and sporadic load, which is precisely the typical profile of IoT traffic.

7.5 Random Access: from ALOHA to CSMA

In pure **ALOHA**, a device transmits as soon as it has data, with no prior check of the state of the channel. If it receives no acknowledgement, it assumes a collision occurred and retransmits after a random interval, precisely to avoid repeating the collision. It is the simplest protocol possible and the one that allows the cheapest device, but its maximum efficiency is low, around 18% of channel capacity. Slotted ALOHA doubles that efficiency by forcing

transmissions to begin at predefined instants. This is the access principle of LoRaWAN and Sigfox, which illustrates the underlying trade-off well: mediocre channel efficiency is accepted in exchange for an extraordinarily simple and cheap terminal.

CSMA (*Carrier Sense Multiple Access*) adds one simple step to random access: listen to the channel before transmitting. If the channel is busy, the device defers transmission; if it is free, it transmits. This single change drastically reduces collisions compared with ALOHA, and is therefore the basis of the vast majority of random access protocols in use.

Collisions do not disappear, however, because of the **propagation delay**: two nodes may find the channel free at practically the same moment and both transmit, because the signal of one has not yet reached the other at the moment of the decision. The interval during which this is possible is called the **vulnerability period**, and grows with the distance between the nodes. An important, and at times counterintuitive, result follows: CSMA degrades as the network grows in physical extent, and not only in number of nodes.

7.6 CSMA/CD and CSMA/CA

In **CSMA/CD** (*Collision Detection*), the device keeps listening to the medium while transmitting, and aborts the transmission as soon as it detects a collision. Aborting early avoids wasting the channel on a frame already known to be lost, which significantly increases efficiency. After the collision, the device waits a random time, given by a binary exponential backoff algorithm, before trying again. It is the historical protocol of shared Ethernet; in modern switched Ethernet networks, where every link is point to point and full-duplex, it is no longer needed.

On a radio medium, however, collision detection is impractical: the transmitter cannot listen to the medium while transmitting, because its own signal saturates the receiver. **CSMA/CA** (*Collision Avoidance*) therefore replaces

detection with **prevention**, seeking to reduce the probability of collision instead of detecting one after it happens. To that end, even when it finds the channel free, the device waits a random interval before transmitting, desynchronising nodes that were waiting for the same moment. Since the sender cannot tell by itself whether the frame arrived, the receiver must **acknowledge** it explicitly; the absence of an acknowledgement is read as a collision. On failure, the random waiting interval is doubled at each new attempt, so as to relieve the channel when load is high.

CSMA/CA is the protocol of Wi-Fi and of IEEE 802.15.4, the standard underlying ZigBee and 6LoWPAN, and will therefore reappear in several lessons of this course.

7.7 The Hidden Terminal Problem

There is one situation in which channel listening fails completely, and it is worth analysing carefully because it exposes the fundamental limit of CSMA.

Consider three nodes in a line: A and C are both within range of B, but too far apart to hear each other. A listens to the channel, finds it free (because it does not hear C) and transmits to B. C does exactly the same, at the same time. The two transmissions collide at B, and neither A nor C has any way of knowing.

The listening mechanism fails here precisely because the decision is taken with local information, at the sender, while the collision happens at the receiver. The classic solution is a prior exchange of short medium reservation messages: the sender issues a request to send and the receiver replies granting it. Because the authorisation comes from the **receiver**, it reaches every node close to it, including those hidden from the sender, which thus learn that they must stay silent. The cost is the extra traffic of the reservation messages themselves, which is why the mechanism is typically enabled only for frames above a given size.

7.8 Comparison and Implications for IoT

Under **high load**, channel partitioning and taking turns are superior: they waste no capacity on collisions and guarantee bounded delay. Under **low and sporadic load**, random access is superior: an isolated node uses the whole channel immediately, without waiting for its turn or its slot.

IoT traffic is typically sporadic and small, which explains the predominance of CSMA and ALOHA variants in short range technologies and in unlicensed long range technologies. Cellular networks, which have to serve human and machine traffic simultaneously with service guarantees, adopt the opposite logic, of centrally scheduled access, the topic of the following chapter.

7.9 Addressing, Ethernet and Wireless Local Networks

The link layer address, also called physical address or **MAC address**, identifies devices connected to the same network. It is technology dependent and set by the manufacturer of the network interface; on local networks it typically has 48 bits, written in hexadecimal. The first bits identify the manufacturer, and two bits signal whether the address is unicast or multicast and whether it is of global or local scope. In sensor networks, the size of the address matters in its own right: IEEE 802.15.4 admits short 16 bit addresses precisely to reduce the weight of the header in very small frames.

Ethernet (IEEE 802.3) has dominated the local network market since the 1970s, despite technically superior alternatives having existed. It is simple and cheap: it establishes no connections and offers no reliability, delegating those functions to the upper layers. The frame contains the destination address, the source address, a type field indicating the protocol of the layer above, the data and an error check code. The Ethernet switch stores and forwards frames according to a forwarding table, transparently to the end devices. The table is built by **learning**: for each frame received, the switch associates the source

address with the interface it arrived on. If the destination is not in the table, the frame is sent to every interface except the one it came from; if it is, the frame is sent only to the corresponding interface.

In **wireless local networks**, each station (STA) associates with an access point (AP), and the set formed by an access point and its stations is called a BSS (*Basic Service Set*). Access points announce the network periodically, including its identifier (SSID), and stations look for the identifiers they know. The IEEE 802.11 standard, in its first version of 1997, operates at 2.4 GHz and 5 GHz; successive amendments increased throughput through bandwidth aggregation and multiple antennas (MIMO). Medium access is done by CSMA/CA, with acknowledgement of every frame, and the frame includes duration and quality of service fields to manage the reservation of the medium.

8 Cellular Mobile Networks

The technologies covered so far reach at most a few tens or hundreds of metres, and assume practically static devices. Mobile networks solve two further problems: covering entire territories with limited spectrum, and keeping communication alive while the device moves. The answer to both is the same structural idea, which gives the technology its name. This architecture is then reused by NB-IoT and LTE-M to give connectivity to IoT devices scattered across an entire region, which is why it deserves detailed treatment in an IoT course.

8.1 The Cellular Concept

Instead of a single high power transmitter covering the whole area, the territory is divided into **cells**, each served by a base station of moderate power. Because the signal attenuates with distance, the same frequency can be **reused** in another cell far enough away, without significant interference.

Frequency reuse is what makes it possible to serve millions of users with an amount of spectrum that would otherwise reach only a few hundred.

The design consequence is notable: total system capacity grows by **reducing** cell size, and not by increasing transmission power. From this follows the progressive densification of networks, with macrocells for wide coverage and increasingly small cells where demand is intense. Smaller cells also mean shorter distance to the device, hence lower transmission power demanded of the terminal, which has a direct energy impact on IoT devices. On the other hand, the smaller the cells, the more frequent the cell transitions as the device moves.

8.2 Architecture of a Cellular Network

A cellular network is organised into three blocks. The **user equipment** (UE) is the terminal, which may be a mobile phone, an industrial module or a sensor with an integrated cellular module. The **radio access network** (RAN) is the set of base stations, called eNodeB in 4G and gNodeB in 5G, responsible for the radio interface and for access scheduling. The **core network** brings together the functions of authentication, mobility management, billing and connection to external networks, notably to the Internet.

This separation between radio access and core is what allows the same core to serve different access technologies, and allows each to evolve at its own pace. The subscriber identity resides in a dedicated module, the SIM card, today often in the form of a soldered eSIM, better suited to sealed IoT devices installed in the field. Network management belongs entirely to the operator, which brings service and coverage guarantees, but also contractual dependency and a recurring cost per device.

8.3 Medium Access in Cellular Networks

Unlike Wi-Fi, the cellular network does not let terminals contend for the medium: access is **centrally scheduled** by the base station. The base station knows the demand of each terminal and the channel quality of each one, and distributes time and frequency resources among them. In the downlink, the base station is the only transmitter, so there are no collisions; in the uplink, each terminal transmits only when and where it is told to.

This model allows quality of service guarantees that random access cannot offer: bounded latency, reserved throughput and differentiated priorities. It also allows tight power control, with the terminal transmitting only the power needed, which reduces interference and saves battery. The price is complexity: the synchronisation and control signalling required make the terminal more expensive and more power hungry than a simple random access radio. It is exactly this trade-off that separates cellular solutions from unlicensed LPWAN.

One problem remains, however: a terminal that has no resources assigned yet needs to request them, but has no scheduled means of doing so. The solution is to reserve a **random access channel**, in which terminals contend by sending a preamble chosen from a limited set. If two terminals choose the same preamble at the same instant, a collision occurs and both repeat the procedure after a random interval. A cellular network therefore combines the two access families: random only for the initial request, scheduled for all subsequent traffic. In massive IoT scenarios, with tens of thousands of devices waking up at once, it is precisely this channel that becomes the bottleneck of the system.

8.4 Multiple Access Techniques and Duplexing

The evolution of mobile generations can be read as a succession of multiple access techniques. **1G** was analogue and used frequency division (FDMA), with

each call occupying a dedicated channel. **2G**, with GSM, was digital and combined frequency and time division (FDMA and TDMA), with several users sharing the same carrier in distinct slots. **3G**, with UMTS, adopted code division (WCDMA), with all users sharing the same wide band, separated by orthogonal codes. **4G**, with LTE, adopted OFDMA in the downlink and, in the uplink, a variant with lower peak to average power, precisely to spare the terminal battery. **5G**, with NR, uses OFDMA in both directions, with configurable subcarrier spacing, which allows the same technology to serve traffic from very low latency to broadband.

The reading of this evolution is clear and pertinent for IoT: each generation sought finer granularity in resource assignment, to accommodate increasingly heterogeneous traffic types, machine traffic among them.

Besides dividing the channel among users, the two directions of communication must be separated. In **frequency division duplexing** (FDD), each direction uses a distinct band, which allows simultaneous transmission in both directions, at the cost of paired spectrum. In **time division duplexing** (TDD), both directions share the same band, alternating in time, which removes the need for paired spectrum and allows the split to be matched to the asymmetry of the traffic. For IoT, TDD has a further advantage: since traffic is strongly asymmetric, with the device sending far more than it receives, the split can be adjusted accordingly.

8.5 Generations and Mobility

In terms of services, **1G** offered analogue voice, with no confidentiality and no data. **2G** brought digital voice, short messages and the first data services, with GPRS and EDGE introducing packet switching into the mobile network. **3G** brought mobile broadband with throughput that made Internet access from the terminal viable. **4G** (LTE) made the network fully packet switched, with voice carried over IP, and throughput of the order of hundreds of Mbit/s. **5G** offers

throughput of the order of Gbit/s, very low latency and, for the first time, native support for massive machine communication. Note that only from 4G onwards, with NB-IoT and LTE-M introduced in 2016, did cellular standardisation begin to treat IoT as a first class use case, rather than as conventional data traffic.

As to mobility, as the terminal moves, the signal of the current cell degrades and that of a neighbouring cell improves, making it necessary to transfer the connection. That process is called **handover** and is decided from measurements the terminal itself reports periodically to the network. Handover may be abrupt, with the previous connection interrupted before the new one is established, or soft, with the two connections momentarily active in parallel. The network also keeps a record of the approximate location of each terminal, so that it can reach it when there is traffic addressed to it. **Roaming** extends this mechanism to networks of other operators, based on agreements that let the terminal be authenticated by its home network while being served by a visited one.

In IoT, these functions carry their own weight. Many devices are fixed and need no handover, but international deployment of equipment often justifies SIMs with permanent roaming. For this reason, NB-IoT does without handover in active connection, while LTE-M keeps it, precisely to serve use cases with real mobility.

8.6 5G and Machine Type Use Cases

5G is organised explicitly around three families of use cases. **eMBB** (*enhanced Mobile Broadband*) is high throughput mobile broadband, the natural continuation of the evolution of previous generations. **URLLC** (*Ultra-Reliable Low-Latency Communications*) is aimed at communications with very low latency and very high reliability, for industrial automation, autonomous vehicles and critical applications. **mMTC** (*massive Machine-Type Communications*) is aimed at massive

machine communication, sized for densities of the order of one million devices per square kilometre.

The three families demand mutually contradictory optimisations: throughput, latency and device density cannot all be maximised at once. This is the same fundamental trade-off that section 13.3 of these notes states between distinct technologies, now formulated within a single technology. The 5G answer is **network slicing**: instantiating several logical networks over the same physical infrastructure, each configured and sized for one traffic profile. For IoT, the relevant category is mMTC, and it is there that the evolution of NB-IoT and LTE-M integrated into 5G belongs.

9 From Communication Networks to IoT Technologies

Everything covered in the previous chapters describes networks designed for people: high throughput, guaranteed power supply and tolerance for complex protocols. IoT technologies solve the same problems, that is, medium access, addressing, encapsulation and routing, but under radically different constraints. This chapter makes that bridge explicit, before the rest of these notes moves on to the overview of concrete technologies.

9.1 The Layered Model in an IoT Solution

It is worth mapping the layered model to the technologies studied in this course. At the **physical and link** layers are IEEE 802.15.4, Bluetooth Low Energy, LoRa and NB-IoT. This is where range and consumption are decided, and this is where the medium access control studied in chapter 7 lives. At the **network** layer is IPv6 compressed by 6LoWPAN, with mesh routing by RPL, topics of Lessons 08, 09 and 10. At the **transport** layer, UDP predominates over TCP, being lighter and requiring no connection establishment, which suits

devices that wake up, send and go back to sleep. At the **session and application** layers are CoAP, MQTT and AMQP, designed for small headers and for the publish and subscribe pattern, topics of Lessons 11 and 12.

The gateway is the point where the lower layers change: below it, a short range technology; above it, conventional IP over the access network of the building or of the operator. This translation is exactly the function of the M2M gateway described in chapter 3, now with the technical vocabulary needed to describe it precisely.

9.2 Medium Access and Energy Consumption

There is a consequence of chapter 7 that deserves separate treatment, because it is the dominant design criterion in IoT and is frequently underestimated.

On a battery powered device, the radio is typically the component that consumes most energy, and it consumes it both transmitting and **listening**. Listening to the channel is therefore not free. CSMA forces the device to keep the radio on before every transmission, and idle listening is one of the main sources of energy waste in a sensor network. On top of this, every collision means energy spent on a lost transmission, to which the energy of the subsequent retransmission is added.

From this follows the renewed interest in TDMA for IoT, despite its being the oldest of the techniques studied: knowing its slot in advance lets the device switch the radio off completely for the rest of the time. That is precisely the principle of 6TiSCH, studied in Lesson 10, which combines time slots with frequency hopping over IEEE 802.15.4.

More generally, the determining parameter is the **duty cycle**, that is, the fraction of time the radio is active. Lifetimes of several years are only possible with duty cycles well below 1%. The design consequence is clear, and worth

retaining: in IoT, the medium access protocol is not chosen for the throughput it offers, but for the time it lets the radio stay off.

9.3 Cellular Networks for IoT: NB-IoT and LTE-M

NB-IoT is a very narrowband technology, occupying about 180 kHz, which can be deployed inside an LTE carrier, in its guard band or standalone. It offers throughput of the order of a few tens of kbit/s and a coverage improvement of about 20 dB over earlier cellular technologies, which lets it reach basements and deep indoor areas where a conventional mobile phone would have no service. It deliberately gives up mobility in active connection and voice, in order to simplify the terminal and reduce consumption and cost.

LTE-M, also called eMTC, occupies about 1.4 MHz and offers throughput of the order of Mbit/s, supporting mobility with handover and voice over IP. The choice between the two comes down, essentially, to mobility and throughput (LTE-M) against coverage, cost and battery life (NB-IoT).

Both rely on two mechanisms that are decisive for battery life: a power saving mode in which the device stays registered on the network but unreachable for long periods, and extended discontinuous reception, which spaces out the instants at which the device listens to the network. These mechanisms are the transposition, into the cellular world, of the reduced duty cycle principle stated in the previous section: the device stays on the network without keeping the radio on.

9.4 Licensed Cellular and Unlicensed LPWAN

The comparison between the two large families of long range solutions can now be made in precise technical terms. Cellular solutions operate in licensed spectrum, with scheduled access, service guarantees and operator infrastructure, but imply a recurring cost per device. Unlicensed LPWAN solutions, such as LoRa and Sigfox, operate in shared spectrum, with random

access and without guarantees, but allow owned infrastructure and need no subscription.

The essential technical difference lies in medium access, and almost all the others follow from it: centralised scheduling on one side, pure contention on the other, with direct consequences for latency, predictability and scalability. Neither option is universally better. It is the nature of the application, the device density and the business model that determine the choice, a topic revisited in detail in Lesson 03.

9.5 Summary

A network is organised into edge, access and core, and carries information by packet switching, with latency breaking down into processing, queuing, transmission and propagation. Layered organisation lets the same application run over any access technology, but each layer adds headers that, in IoT, may exceed the useful information.

On shared media, the central problem is medium access, and the answers fall into channel partitioning, taking turns and random access. Random access suits sporadic traffic and underpins Wi-Fi, IEEE 802.15.4 and unlicensed LPWAN; scheduled access underpins cellular networks and their service guarantees. Mobile networks solve coverage and mobility through the cellular concept, and reserve random access only for the initial resource request.

Above all, it should be retained that, in IoT, the dominant criterion when choosing a medium access protocol is not throughput, but the fraction of time it allows the radio to stay off. With this framing, the following chapters go through the concrete technologies, each corresponding to a distinct trade-off between these axes.

10 Framing: Scale, Requirements and Spectrum

Before comparing concrete technologies, it is important to fix the criteria that make that comparison possible: the spatial scale of the network, the requirements specific to IoT communications and the regime governing use of the radio spectrum.

10.1 Classification of Networks by Spatial Scope

Computer networks are classified, among other criteria, by the physical extent they cover. That extent is not a descriptive detail: it directly constrains which communication technologies are applicable, because the range of a radio technology follows from the available power, the frequency used and the regulatory restrictions that apply to it.

At the smallest scale are near field networks (NFC) and body area networks (BAN), with ranges from centimetres to a few metres. Next come personal area networks (PAN), home networks (HAN) and local networks, wired and wireless (LAN and WLAN), typical of a dwelling or an office building. At larger scales are campus networks (CAN), metropolitan networks (MAN), including municipal wireless networks, and wide area networks (WAN), up to the Internet itself.

An IoT solution rarely lives at a single one of these scales. The common case combines several: sensors interconnected by a short range personal network, aggregated at a local gateway, which in turn communicates with cloud services over a wide area network. Understanding this stratification is the first step towards choosing technologies on an informed basis.

10.2 IoT Communication Requirements

IoT communications depart, along several axes, from what is demanded of a cellular network designed for human traffic. Cost per device and per connection has to be low, because the number of nodes is high and grows with the size of the installation. Energy consumption has to be low, with battery lifetimes often measured in years, since many devices end up installed in places that are hard to reach. The number of simultaneous connections per cell or per gateway is far higher than in a voice network. Throughput requirements are variable, but almost always modest. And the devices themselves are small, with limited processing and storage capability, which calls for simple architectures and protocols.

This list does not describe a single application, but a space of applications with very different requirements. The bandwidth needed by a temperature sensor reporting one value per hour has nothing to do with that of a surveillance camera. The range required separates applications confined to a building from applications spread over an agricultural region. Energy autonomy is decisive in battery powered devices and irrelevant in devices connected to mains power. Quality of service requirements vary with the criticality of the application. It is precisely this diversity that justifies making the radio spectrum available in several frequency bands, suited to different use cases, and that explains why no universal technology exists.

10.3 Licensed and Unlicensed Spectrum

The spectrum used by a technology may be licensed or unlicensed, and the difference has relevant practical consequences.

Licensed spectrum is assigned to operators under licence. It offers better interference management, because the number of authorised transmitters is controlled, and allows greater network security and higher reliability, with

contractually binding quality of service guarantees. It has the further advantage of allowing reuse of the cellular infrastructure already installed and of the device ecosystem that comes with it, which reduces the effort of deploying M2M and IoT solutions over existing mobile networks.

Unlicensed spectrum requires no licence fees, which significantly reduces the cost of operation. In exchange, it is subject to regulatory limits on radiated power (EIRP restrictions) and offers no guarantee of quality of service, since any device meeting the technical conditions set in the regulation may access the same band. Short range and delay tolerant applications are the typical use cases for this regime.

The choice between the two regimes amounts, in essence, to a trade-off between cost of operation and service guarantees. A solution for metering energy consumption in an urban area may accept the uncertainty of unlicensed spectrum; a safety critical application hardly will.

10.4 Overview of Technical Solutions

The set of technologies available for IoT is organised into families. Among cellular M2M solutions, operated in licensed spectrum, are NB-IoT, eMTC and LTE V2X, the last of these aimed at vehicular communications. Among non cellular long range solutions, in unlicensed spectrum, are LoRa, Sigfox, Ingenu and the Weightless family, in the N, P and W variants. Among short range solutions are Bluetooth and Bluetooth Low Energy, ZigBee, Z-Wave, Wi-Fi, NFC and RFID. Wi-Fi also has IoT oriented variants, such as IEEE 802.11ah, designed for greater range and lower consumption, and IEEE 802.11p, intended for vehicular communications. Where fixed infrastructure is available, wired solutions (Ethernet and Powerline) remain valid options.

From an architectural point of view, these technologies are distributed across layers. At the base are the sensors and actuators, connected by short range technologies or by cable. These devices concentrate at gateways, industrial or

domestic, which aggregate the traffic of the local sensor network. The gateway then connects to the transport infrastructure (cellular networks, optical fibre, cable or satellite) and, at the top, are the cloud storage services and the analytics layers that process the collected data.

11 Wired Solutions

Not every IoT solution uses wireless communication. Where fixed infrastructure exists and the devices have no mobility or power supply constraints, wired solutions retain clear advantages.

11.1 Ethernet (IEEE 802.3)

Ethernet is a low cost, highly reliable technology, with bandwidth performance far above that of the wireless alternatives. It also allows devices to be powered through the cable itself, using Power over Ethernet, although this requires additional circuitry and components in the equipment.

The limitations are equally clear. It requires a permanent cable connection, which restricts mobility and makes installation expensive in already constructed buildings. Current draw is high and continuous, typically of the order of 200 to 300 mA, which makes it incompatible with battery powered devices. And the connector is bulky, which limits its integration into small devices. In IoT solutions, Ethernet therefore appears mostly on the gateway side, and not on the sensor side.

11.2 PowerLAN

PowerLAN consists of carrying Ethernet frames over the existing electrical network, taking advantage of the cabling already installed in the building. It should not be confused with Power over Ethernet: in that case, the data network carries power; in PowerLAN, the electrical network carries data.

There is a wide diversity of models and characteristics on the market, some of which integrate Wi-Fi so as to act simultaneously as wireless coverage extenders. Advertised speeds can reach 2 Gbit/s, but effective performance depends heavily on the distance between points and on the quality of the electrical installation. Electrical noise introduced by other equipment connected to the same network is a real threat to the operation of the link, which makes this solution less predictable than conventional Ethernet.

12 Radio Frequency Identification: RFID and NFC

RFID and NFC occupy the lower end of the range scale. They do not constitute networks in the conventional sense, but they solve the problem of identifying objects and of interaction by proximity, functions that recur in IoT solutions.

12.1 RFID: Concept and Operation

RFID uses radio waves to read information stored in a tag attached to an object. The first applications date back to the 1940s, and the first patent associated with the technology was granted to Charles Walton in 1983. The technology allows objects to be identified, metadata to be recorded or a target to be controlled individually, and has the particular feature of not requiring direct line of sight between the reader and the tag, unlike the barcode, which performs an equivalent function but requires optical reading.

An RFID system is made up of three elements: a tag, a reader (also called interrogator) and an antenna. The tag contains a microchip, which stores and processes the information, and an antenna, which receives and transmits the signal; it typically holds a unique serial number of the object it is attached to. The reader is normally connected to a host computer or to a network, where the collected information is processed.

Operation rests on electromagnetic induction. The coils of the reader create a magnetic field when carrying electrical current. The antenna of the tag receives that excitation, which allows it to reply. In passive tags, the energy received in the excitation is used to power the circuit itself, which does away with a battery. The tag then replies with data, which may be a simple identifier or the answer to a challenge sent by the interrogator. The technology operates in several frequency ranges: LF between 125 and 134 kHz, HF at 13.56 MHz, and UHF at 433 MHz and between 860 and 915 MHz.

12.2 Passive and Active Tags

The distinction between passive and active tags determines which applications are possible.

A passive tag holds no battery and uses the energy of the radio wave from the interrogator to return the stored information. It therefore works without maintenance, with a useful life exceeding twenty years. It is cheap to manufacture and very small, with dimensions comparable to a grain of rice, which opens up an almost unlimited range of applications in consumer goods. In exchange, it can only be read at very short distances, which strongly restricts some applications, and it hardly allows sensors requiring electrical power to be integrated.

An active tag has a battery of its own. This gives it greater read range and allows additional functionality to be integrated, including sensing, at the cost of higher price, larger size and the need for periodic maintenance.

12.3 NFC: Concept and Specifications

NFC is a set of communication protocols between electronic devices at very short distance, typically between 0 and 2 cm, reaching 10 cm in certain configurations. It is a subset of RFID technology, designed specifically for devices in close proximity. It uses inductive electromagnetic fields, unlike

Bluetooth and Wi-Fi, which rest on radio transmission proper. It offers a low throughput connection with simple setup, which makes it suitable for initiating, or bootstrapping, more capable wireless connections. It allows communication between two powered devices, both active, or between a powered device and a passive element with no power source of its own.

From a technical point of view, NFC operates at 13.56 MHz, in an ISM band available globally and unlicensed, over the air interface defined in standard ISO/IEC 18000-3. It supports throughputs of 106, 212 and 424 kbit/s. The theoretical operating distance with standardised compact antennas reaches 20 cm, but the practical distance is around 4 cm.

The technology was approved as an ISO/IEC standard in December 2003 and later as an ECMA standard. The reference standards are ISO/IEC 18092, corresponding to ECMA-340, which defines the NFCIP-1 interface and protocol, and ISO/IEC 21481, corresponding to ECMA-352, which defines NFCIP-2. NFC also incorporates standard ISO/IEC 14443. On top of these standards, the NFC Forum defined a common data format, called NFC Data Exchange Format (NDEF), which ensures interoperability between implementations from different manufacturers.

12.4 Modes of Operation, Tag Types and Applications

NFC defines three modes of operation. In reader and writer mode, the active device reads or changes the content of a passive NFC tag. In peer-to-peer mode, standardised by the NFC Forum in 2009, two active devices exchange information with each other. In card emulation mode, the device behaves towards the reader as a conventional contactless card, which underpins payment applications.

The behaviour of each element follows from these modes. An active device, typically a mobile phone with NFC, collects information from tags, exchanges data with other compatible devices and can change the content of the tag

when authorised to do so. A reader continuously emits radio frequency carrier signals and watches the received signals waiting for data. A passive tag has no power source of its own: it absorbs the energy emitted by the reader and replies with modulated information.

As to tag types, types 1 and 2 are based on standard ISO 14443-A, are read and write and may be configured by the user as read only; available memory is 48 or 96 bytes, expandable to 2 KB. Type 3 is based on the Japanese standard JIS X 6319-4, known as Sony FeliCa, with the read and write configuration set at the time of manufacture.

Current applications include smart posters, mobile payment, sharing of small files such as contacts, and peer to peer interaction scenarios. The technology is also used in electronic identity documents and in access cards. The advantages lie in convenience (the interaction amounts to a single tap) and in versatility, with application from bank cards to transport passes and loyalty systems. The limitations are of two kinds: on one hand, dependence on companies adopting the ecosystem, since without integration by merchants the technology does not reach the user; on the other, exposure to security risks, which grows as mobile phones concentrate sensitive information and become attractive targets.

13 Short Range Wireless Technologies

This chapter goes through the technologies that underpin personal and local networks in IoT solutions, and ends with the criteria that guide the choice between them.

13.1 Bluetooth

Bluetooth is a low power wireless technology operating at short range in the 2.4 GHz ISM band. It emerged as a wireless alternative to cable connections

and underpins the creation of personal area networks (PAN). It supports throughput of the order of 1 Mbit/s, suited to data traffic and, in more recent versions, to video traffic. It uses frequency hopping spread spectrum, which increases robustness against interference in a band shared by many other devices. It operates in connected and non connected mode, according to whether an association between devices has been established.

Range depends on the power class of the device. Class 1, with a maximum power of 100 mW (20 dBm), reaches about 100 m. Class 2, with 2.5 mW (4 dBm), reaches about 10 m. Class 3, with 1 mW (0 dBm), stops at about 1 m. Bluetooth 5 quadrupled the range compared with the previous generation, which steered the technology decisively towards IoT scenarios. Bluetooth is revisited in detail in lessons 4 and 5.

13.2 ZigBee and Wi-Fi

ZigBee is a low cost, widely available technology aimed at sensor networks with reduced energy consumption. It defines three distinct roles for the devices making up the network. The coordinator acts as the root of the network and as a bridge to other networks, being responsible for forming the network. The router is an intermediate device that lets data cross the network towards other devices, extending coverage. The end device has limited functionality and communicates only with its parent node, which lets it spend most of the time at rest and save energy. This organisation is revisited in detail in lesson 7.

Wi-Fi covers a range of variants with very different characteristics. IEEE 802.11a operates at 5 GHz, with throughput up to 54 Mbit/s and typical range of 10 m. IEEE 802.11b operates at 2.4 GHz, with throughput up to 11 Mbit/s and typical range of 140 m. IEEE 802.11g operates likewise at 2.4 GHz, with throughput up to 54 Mbit/s and range of 140 m. IEEE 802.11n operates at 2.4 GHz and 5 GHz, with throughput up to 450 Mbit/s and typical range of 250 m.

The high consumption of these variants limits their application in battery powered devices.

IEEE 802.11ah, commercially known as Wi-Fi HaLow, answers this limitation. It operates at 900 MHz, with throughput of the order of 8 Mbit/s, favouring range and consumption over throughput. Its reduced consumption puts it in direct competition with Bluetooth in sensing applications, while offering higher throughput and wider coverage than conventional short range technologies. Operating at lower frequencies further improves penetration through obstacles, which favours coverage inside buildings. It is, in practice, an intermediate point between classical wireless local networks and the long range, low power technologies covered in the following lesson.

13.3 Selection Criteria and Trade-offs

No technology combines low cost, long range and high bandwidth at once. The choice always means giving way on one of the axes, and recognising this early avoids unrealistic specifications. The principle is stated, ironically but accurately, in RFC 1925, known as “The Twelve Networking Truths”: of good, fast and cheap, you may pick only two.

Applied to the technologies covered in this lesson, the principle translates as follows. Short range technologies, such as Bluetooth or ZigBee, favour consumption and cost, sacrificing range. Cellular technologies offer coverage and quality of service, but bring licensing costs and higher consumption. RFID and NFC offer very low cost per tag and need no power supply, at the price of a range of centimetres and the absence of a network proper. Conventional Wi-Fi offers high throughput at the cost of consumption incompatible with prolonged battery life.

To recapitulate: the spatial scale of the network is the first selection criterion; IoT communication requirements depart from those of human communications, and it is that difference which justifies dedicated technologies; the licensing

regime of the spectrum determines the trade-off between cost of operation and service guarantees. Among short range technologies, RFID and NFC serve identification and proximity, Bluetooth serves personal connections, ZigBee serves mesh sensor networks and Wi-Fi serves high throughput at higher consumption. The following lesson covers long range, low power technologies, which answer the scenarios that none of these solutions covers.